Wildfire policy and public lands: integrating scientific understanding with social concerns across landscapes

MP Dombeck, JE Williams, CA Wood - Conservation biology, 2004 - Wiley Online Library

… **wildfire** in the United States and our attempts to control it contain many lessons to guide future policies for **management** … weeds **spread** at a rate of 688,500 ha annually (Bureau of Land …

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A real-time web-based **wildfire simulation** system

R Wu, C Chen, S Ahmad, JM Volk… - IECON 2016-42nd …, 2016 - ieeexplore.ieee.org

… Therefore, a simplified **model** of fire **propagation simulation** was developed in this study. … taken intermittingly by the United States Geological **Survey** (USGS) between 1948 and 2010. …

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Interpreting and using outputs from the Canadian **Forest Fire** Danger Rating System in research applications

BM Wotton - Environmental and ecological statistics, 2009 - Springer

… **review** will expose statistical modellers and other researchers to some of the models used currently in **forest fire management** … quantities such as rate of **spread**, fuel consumption and …

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Automatic **wildfire** detection and **simulation** using optical information from unmanned aerial systems

CW Lum, A Summers, B Carpenter, A Rodriguez… - 2015 - sae.org

… are simply not able to **survey** the amount of area necessary … (UAS) for automated **wildfire** detection. The proposed system … for the detection and **simulation** of fire **propagation** from UAS …

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Spatial-temporal distribution **prediction** of transmission corridor **wildfire** risk based on ARIMA-DBN

E Zhou, L Wang, H Wang, Z Wang… - Frontiers in Forests and …, 2025 - frontiersin.org

… **model** for assessing the spatiotemporal risk of **wildfire** occurrence in transmission corridors, with an emphasis on the role of meteorological factors in short-term **wildfire dynamics**. A …

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Integrated **wildfire** risk assessment: Framework development and application on the Lewis and Clark National Forest in Montana, USA

MP Thompson, J Scott, D Helmbrecht… - … and **management**, 2013 - academic.oup.com

… consider biophysical variables beyond fire **behavior**. We anticipate that this work will further the state of … Below we briefly **review** each step in the context of the Little Belts Assessment. …

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Wildfire management in the United States: the evolution of a policy failure

G Busenberg - **Review** of policy research, 2004 - Wiley Online Library

… **model** of the origins and **propagation** of policy failures. The elements of this policy failure **model** … This article applies the policy failure **model** to explore the evolution of US **wildfire** policy, …

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Prediction of **forest fire spread** rate using UAV images and an LSTM **model** considering the interaction between fire and wind

X Li, H Gao, M Zhang, S Zhang, Z Gao, J Liu, S Sun… - Remote Sensing, 2021 - mdpi.com

… **model** to predict the **spread** rate of the **forest fire**. In order to deeply capture the characteristics of **forest fire spread** by … impact to the process of **forest fire spread** as the input parameters to …

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A Comparative Study of a Deep Reinforcement Learning Solution and Alternative Deep Learning Models for **Wildfire Prediction**.

C Vidal-Silva, R Pizarro… - Applied Sciences …, 2025 - search.ebscohost.com

… dependencies in **wildfire dynamics**. The findings can support the integration of deep learning models into early warning systems, contributing to proactive **wildfire** risk **management**. …

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Trying not to get burned: Understanding homeowners' **wildfire** risk–mitigation behaviors

H Brenkert-Smith, PA Champ, N Flores - Environmental **Management**, 2012 - Springer

… WUI property owners about **wildfire** risks. In this study, we analyze **survey** data from WUI … to the current literature on homeowners and **wildfire** risk–mitigation **behavior**. First, we move …

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LinU-Mamba: Visual Mamba U-Net with Linear Attention to Predict **Wildfire Spread**

HS Andrianarivony, MA Akhloufi - Remote Sensing, 2025 - mdpi.com

… In this study, we develop a deep learning **model** to predict **wildfire spread** using remote … Collectively, these deep models deliver the highest performance reported in the **review**, but the …

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Cloud-based urgent computing for **forest fire spread prediction**

E Fraga, A Cortés, T Margalef, P Hernández… - … Modelling & Software, 2024 - Elsevier

… **Forest fire simulation** have improved significantly, but input data describing fire scenarios are subject to high levels of uncertainty. In this work the two-stage **prediction** … actual **prediction** …

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A comprehensive analysis of the effect of radiation and Sherwood numbers in **wildfire model** with moisture convection and diffusion mechanisms

AB Zhiri, OA Ajala, AO Akindele, P Adegbite… - Journal of Radiation …, 2025 - Elsevier

… The **spread** of forest fires in this study has been modeled using a … **model** offers a valuable tool for researchers to investigate how climate change may influence future **wildfire behavior**…

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A **review** of **forest fire** surveillance technologies: Mobile ad-hoc network routing protocols perspective

FT Al-Dhief, N Sabri, S Fouad, NMA Latiff… - Journal of King Saud …, 2019 - Elsevier

… and developers for various applications such as **forest fire** detection. Forest fires require continuous … This paper gives an extensive **review** of the modern techniques used in the **forest fire** …

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Wildfiredb: An open-source dataset connecting **wildfire spread** with relevant determinants

S Singla, A Mukhopadhyay, M Wilbur, T Diao… - Conference on Neural …, 2021 - cs.ucr.edu

… Our goal is to present a data source that can be used to **model** both fire occurrences and the **spread** of fires. We discretize space and time to create our dataset (we discuss the unit of …

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Mapping regional **forest fire** probability using artificial neural network **model** in a Mediterranean forest ecosystem

O Satir, S Berberoglu, C Donmez - Geomatics, Natural Hazards …, 2016 - Taylor & Francis

… In this paper, the basic feedforward, back-**propagation** ANN described above is used as a regression **model** to estimate fire probability based on input features that include the climate, …

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Coupled atmosphere-**wildland fire** modeling with WRF-Fire version 3.3

J Mandel, JD Beezley… - Geoscientific **Model** …, 2011 - gmd.copernicus.org

… **wildland fire behavior** can be captured by the coupling of a mesoscale weather **model** with a simple 2-D fire **spread model** (… Sullivan, AL: A **review** of **wildland fire spread** modelling, 1990…

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Fire intensity, fire severity and burn severity: a brief **review** and suggested usage

JE Keeley - International journal of **wildland fire**, 2009 - CSIRO Publishing

… This metric is most useful in understanding fire **behavior** in … precise metric varies with **management** needs. Confusion arises … of these terms and suggest a **model** that may help clarify …

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A decision-making framework for **wildfire** suppression

N Petrovic, JM Carlson - International Journal of **Wildland Fire**, 2012 - CSIRO Publishing

… fire **spread model**, an economic cost **model** and population data. Our study uses **model**-… multiple fires using scenarios motivated by the 2003 and 2007 California **wildfire** seasons. …

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The Rothermel surface fire **spread model** and associated developments: A comprehensive explanation

PL Andrews - Gen. Tech. Rep. RMRS-GTR-371. Fort Collins …, 2018 - research.fs.usda.gov

… I hope that this paper will serve as a valuable reference for students of **wildland fire** … Sullivan (2009a,b,c) did a comprehensive **review** of wildland surface fire **spread** models developed …

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